

To verify that *outputfile.apk* is aligned, run the next command:

```
zipalign [-c] [-v] <alignment> outputfile.apk
```

The *<alignment>* parameter is an integer value that defines the byte-alignment boundaries. We always use 4 (for 32-bit alignment) or else it effectively does nothing. The flags used are *[-ff]* to overwrite the existing output file; *[-v]* for verbose output, and *[-c]* to confirm alignment of the given file.

Aligning using Ant

The Ant build script (Android 1.6 onwards) isn't capable of aligning targets for older versions of Android. They need to be aligned manually.

When building in debug mode, Ant signs and aligns the packages automatically. In release mode, Ant aligns packages only if it has enough information to sign the package—i.e., the location of the keystore and the name of the key, in *ant.properties*. The respective properties are named *key.store* and *key.alias*. If these properties are missing, the package will neither be signed nor aligned.

Final words

We hope we've convinced you why unaligned apps are undesirable, and how Zipaligning is a short, sweet solution to this problem. We hope every one of your Android apps will be Zipaligned from now on! Keep experimenting with and enjoying Android! **END** 

References

- [1] developer.android.com
- [2] www.addictivetips.com

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